# InvoiceIQ – Week 1 Internship Report

## Data Cleaning & Preprocessing | Customer Invoices Dataset

Prepared from the dataset supplied for the Week 1 assignment.

## 1. Objective

The objective was to inspect the raw customer invoice data, assess data quality, standardize formats, map available columns to the InvoiceIQ project fields, and create a cleaned dataset with payment-related derived columns.

## 2. Dataset overview

| Metric | Result |
|---|---|
| Rows in original file | 50,000 |
| Columns in original file | 19 |
| Exact duplicate rows removed | 1,161 |
| Rows in cleaned file | 48,839 |
| Open invoices | 9,681 |
| Paid invoices | 39,158 |
| Currencies observed | CAD, USD |

## 3. Cleaning and preprocessing decisions

• Removed 1,161 fully identical duplicate rows, keeping the first occurrence. No non-identical records were removed.

• Trimmed surrounding whitespace in text fields and standardized selected code/category fields to uppercase.

• Parsed clear_date and posting_date as dates; parsed document_create_date, document_create_date.1, due_in_date and baseline_create_date from YYYYMMDD.

• Dropped area_business because all 50,000 source values were missing.

• Retained missing clear_date values because they correspond to open invoices; payment dates were not imputed.

• Retained six missing invoice_id values and used doc_id as the project's primary invoice identifier.

• Flagged due dates earlier than posting dates for mentor review rather than automatically changing them.

• Found no negative or zero values in total_open_amount.

## 4. Project field mapping and limitations

doc_id was used as Invoice ID; cust_number and name_customer form Customer; posting_date is used as the Invoice Date proxy; due_in_date is Due Date; clear_date and isOpen represent payment date/status; total_open_amount is provisionally mapped to Amount; invoice_currency is Currency. The dataset has no country field, so Country is left blank. The amount field's source name is total_open_amount and its business meaning should be confirmed with the mentor before modeling.

## 5. Derived columns

days_to_pay = payment_date – invoice_date. payment_delay_days = payment_date – due_date. paid_late_flag is 1 when a paid invoice was settled after its due date and 0 when paid on or before the due date; it is blank for open invoices. open_paid_flag is Open when isOpen=1 and Paid when isOpen=0. Payment-derived values for open invoices remain missing to avoid inventing future outcomes.

## 6. Deliverables

The submission package contains Cleaned_Invoices.csv, Data_Dictionary.xlsx, Cleaning_Log.xlsx, Week1_Data_Cleaning.py and this report. The Python script is reusable and includes run instructions in its header.

## 7. What I learned

I practiced loading structured data with pandas, checking missing values and duplicates, converting date formats, standardizing categorical fields, documenting data-quality decisions, mapping source columns to business requirements, and creating derived payment features. I also learned not to fill unknown payment dates or infer unavailable country information.

Review note: Confirm with the mentor whether posting_date is the intended invoice date and whether total_open_amount represents the invoice amount required by the project. Country is unavailable in this dataset.